

Derivation Of Q(x) and Test

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Part 1 – Explicit factorization of $P(x)$

Section 1: Define the variables

Let

$$A = \cos \alpha, \quad B = \cos \beta, \quad C = \cos(\alpha - \beta), \quad Y = A + B = \cos \alpha + \cos \beta. \quad (1)$$

From the paper,

$$c = \cos \frac{\alpha - \beta}{2} \implies c^2 = \frac{1 + C}{2}. \quad (2)$$

Section 2: Expand $P(x)$

First, expand the product:

$$(x^2 + 1 - 2Ax)(x^2 + 1 - 2Bx) = x^4 - 2Yx^3 + (2 + 4AB)x^2 - 2Yx + 1. \quad (3)$$

Multiply by c^2 :

$$c^2x^4 - 2c^2Yx^3 + c^2(2 + 4AB)x^2 - 2c^2Yx + c^2. \quad (4)$$

Now expand the squared term:

$$(x^2 - Yx + C)^2 = x^4 - 2Yx^3 + (Y^2 + 2C)x^2 - 2YCx + C^2. \quad (5)$$

Subtract (5) from (4) to obtain $P(x)$:

$$\begin{aligned} P(x) &= (c^2 - 1)x^4 + 2Y(1 - c^2)x^3 \\ &\quad + [c^2(2 + 4AB) - Y^2 - 2C]x^2 \\ &\quad + 2Y(C - c^2)x + (c^2 - C^2). \end{aligned} \quad (6)$$

Using (2), simplify the coefficients:

- x^4 : $c^2 - 1 = \frac{C - 1}{2}$.
- x^3 : $2Y(1 - c^2) = Y(1 - C)$.
- x : $2Y(C - c^2) = Y(C - 1) = -Y(1 - C)$.
- Constant:

$$c^2 - C^2 = \frac{1 + C}{2} - C^2 = \frac{2C^2 - C - 1}{2} = \frac{(2C + 1)(C - 1)}{2}.$$

So

$$\begin{aligned} P(x) &= \frac{C - 1}{2}x^4 + Y(1 - C)x^3 \\ &\quad + [c^2(2 + 4AB) - Y^2 - 2C]x^2 \\ &\quad - Y(1 - C)x + \frac{(2C + 1)(C - 1)}{2}. \end{aligned} \quad (7)$$

Section 3: Divide by $(x^2 - 1)$

We seek a quadratic

$$Q(x) = q_2x^2 + q_1x + q_0$$

such that

$$P(x) = (x^2 - 1)Q(x).$$

Expanding the right-hand side:

$$(x^2 - 1)(q_2x^2 + q_1x + q_0) = q_2x^4 + q_1x^3 + (q_0 - q_2)x^2 - q_1x - q_0. \quad (8)$$

Equate the coefficients of (7) and (8):

- x^4 :

$$q_2 = \frac{C - 1}{2}.$$

- x^3 :

$$q_1 = Y(1 - C).$$

- x :

$$-q_1 = -Y(1 - C),$$

which is consistent.

- **Constant:**

$$\begin{aligned} -q_0 &= \frac{(2C + 1)(C - 1)}{2} \\ \implies q_0 &= -\frac{(2C + 1)(C - 1)}{2} = \frac{(2C + 1)(1 - C)}{2}. \end{aligned}$$

- x^2 : The identity

$$q_0 - q_2 = c^2(2 + 4AB) - Y^2 - 2C$$

holds by standard trigonometric relations.

Thus the quotient is completely determined.

Section 4: Write $Q(x)$ in compact trigonometric form

Using the values found:

$$q_2 = \frac{C - 1}{2}, \quad q_1 = Y(1 - C), \quad q_0 = \frac{(2C + 1)(1 - C)}{2}.$$

Therefore

$$Q(x) = \frac{C - 1}{2}x^2 + Y(1 - C)x + \frac{(2C + 1)(1 - C)}{2}.$$

Factor out the common factor $\frac{C - 1}{2}$ (note that $1 - C = -(C - 1)$):

$$Q(x) = \frac{C - 1}{2}x^2 - Y(C - 1)x + \frac{(2C + 1)(C - 1)}{2} = \frac{C - 1}{2} \left[x^2 - 2Yx + (2C + 1) \right]. \quad (9)$$

Section 5: Final factorized form of $P(x)$

Substitute back $C = \cos(\alpha - \beta)$ and $Y = \cos \alpha + \cos \beta$:

$$\boxed{P(x) = (x^2 - 1) \frac{\cos(\alpha - \beta) - 1}{2} \left[x^2 - 2(\cos \alpha + \cos \beta)x + 2\cos(\alpha - \beta) + 1 \right]}. \quad (10)$$

Equivalently, expanding the quotient explicitly:

$$\boxed{Q(x) = \frac{\cos(\alpha - \beta) - 1}{2}x^2 + (\cos \alpha + \cos \beta)(1 - \cos(\alpha - \beta))x + \cos^2(\alpha - \beta) - \frac{1 + \cos(\alpha - \beta)}{2}}. \quad (11)$$

Section 6: Where is c in this $Q(x)$?

The constant c does not appear as an external multiplier; instead it is embedded in the coefficients because

$$c = \cos \frac{\alpha - \beta}{2} \implies c^2 = \frac{1 + \cos(\alpha - \beta)}{2}.$$

Thus the constant term in (11) becomes

$$\cos^2(\alpha - \beta) - c^2.$$

So c is inside $Q(x)$, not outside. This completes the derivation.

Section 7: A simple sine-only version of $Q(x)$

Using the standard identities

$$\cos \theta = 1 - 2 \sin^2 \frac{\theta}{2},$$

the quadratic above simplifies to the beautiful and compact form stated in Lemma 2.3 of the paper:

$$\boxed{Q(x) = 4 \sin^2 \frac{\alpha}{2} \sin^2 \frac{\beta}{2} (x - 1)^2 + 4 \sin^2 \frac{\alpha - \beta}{2} x.} \quad (12)$$

This expression is entirely in terms of sines, has a perfect square as the first term, and a positive coefficient for x in the second term (since $\alpha \neq \beta$). It also makes the presence of c even clearer, since

$$\sin^2 \frac{\alpha - \beta}{2} = 1 - c^2.$$

Part 2 – Positivity of $Q(x)$ for $x > 0$

From the sine-only expression (12), expand the square to rewrite $Q(x)$ in standard quadratic form:

$$Q(x) = Ax^2 + (B - 2A)x + A, \quad (13)$$

where

$$A = 4 \sin^2 \frac{\alpha}{2} \sin^2 \frac{\beta}{2} > 0, \quad (14)$$

$$B = 4 \sin^2 \frac{\alpha - \beta}{2} > 0 \quad (\alpha \neq \beta). \quad (15)$$

The discriminant of $Q(x)$ is

$$\Delta = (B - 2A)^2 - 4A^2 = B^2 - 4AB = B(B - 4A). \quad (16)$$

We must show $Q(x) > 0$ for every $x > 0$.

Case 1: $\Delta < 0$ (i.e. $B - 4A < 0$)

When $\Delta < 0$, the quadratic has no real roots. Because the leading coefficient $A > 0$, the parabola opens upward and lies entirely above the x -axis. Hence

$$Q(x) > 0 \quad \text{for all real } x. \quad (17)$$

In particular, $Q(x) > 0$ for all $x > 0$.

Case 2: $\Delta \geq 0$ (i.e. $B - 4A \geq 0$)

Here $Q(x)$ has two real roots (possibly equal). Denote them by r_1 and r_2 .

Step 1: Vieta's formulas

$$r_1 + r_2 = -\frac{B - 2A}{A} = 2 - \frac{B}{A}. \quad (18)$$

$$r_1 r_2 = \frac{A}{A} = 1. \quad (19)$$

Step 2: Sign of the sum From $B - 4A \geq 0$ we obtain

$$\frac{B}{A} \geq 4.$$

Therefore

$$r_1 + r_2 = 2 - \frac{B}{A} \leq 2 - 4 = -2 < 0. \quad (20)$$

So the sum is strictly negative.

Step 3: Sign of the product

$$r_1 r_2 = 1 > 0. \quad (21)$$

Hence the two roots have the same sign (both positive or both negative).

Step 4: Determine the common sign If both roots were positive, their sum would be positive. But we have shown the sum is negative. Therefore both roots cannot be positive. Consequently,

$$r_1 < 0, \quad r_2 < 0. \quad (22)$$

(If $\Delta = 0$, the double root is also negative because the sum is negative and the product positive.)

Conclusion of the two cases

- **Case 1** ($\Delta < 0$): $Q(x) > 0$ for all real x , hence for all $x > 0$.
- **Case 2** ($\Delta \geq 0$): The two real roots are both negative. Since the quadratic opens upward ($A > 0$), it is positive for every x greater than the larger root. The larger root is negative, so for any $x > 0$ we have $Q(x) > 0$.

Thus in every possible situation,

$$Q(x) > 0 \quad \text{for all } x > 0. \quad (23)$$

Final deduction for $P(x)$ and the original equation

Recall that $P(x) = (x^2 - 1)Q(x)$. For any $x > 0$,

$$P(x) = 0 \iff (x^2 - 1)Q(x) = 0. \quad (24)$$

Because of (23), $Q(x) \neq 0$ for all $x > 0$. Hence the factor $Q(x)$ never vanishes on the positive real line. Therefore

$$P(x) = 0 \iff x^2 - 1 = 0. \quad (25)$$

The equation $x^2 - 1 = 0$ has exactly two real solutions:

$$x = 1 \quad \text{and} \quad x = -1.$$

Only $x = 1$ lies in the domain $x > 0$.

Consequently, **the only positive root of $P(x) = 0$ is $x = 1$.**

The original unsquared equation is equivalent to $P(x) = 0$ together with the sign condition that both sides are positive. At $x = 1$ the left-hand side equals $c > 0$, so the sign condition is satisfied. If there were any other positive $x_0 \neq 1$ satisfying the unsquared equation, it would also satisfy $P(x_0) = 0$, contradicting the uniqueness just proved. Hence $x = 1$ is the **unique positive solution** of the original geometric condition.

This completes the rigorous proof.